Decentralized Trust-Region Policy Distillation in Networked Multi-Agent Reinforcement Learning: A Theoretical Blueprint

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Abstract

Fully decentralized multi-agent reinforcement learning (MARL) requires each agent to accelerate its own optimization by absorbing knowledge from neighbors, yet naive peer-to-peer policy distillation collapses under distribution mismatch: a neighbor operating in a different local regime supplies destabilizing gradients that overwrite an agent's hard-won competencies. We present a theoretical blueprint for *Decentralized Trust-Region Policy Distillation* (D-TrOPD). Our central conceptual move is to separate two objects that naive formulations conflate: the *policy-improvement trust region*—a bound on each agent's update divergence $\mathrm{KL}(\pi_i^{\mathrm{new}} \,\|\, \pi_i^{\mathrm{old}})$, which alone governs improvement guarantees—and the *distillation trust region*—a radius on the social divergence $\mathrm{KL}(\pi_i \,\|\, \pi_j)$ to a neighbor, which governs how strongly peer knowledge is imported. Constraining only the latter, as in a direct transfer of single-teacher trust-region distillation, does not yield an improvement guarantee. We develop a mechanism that (i) replaces a brittle reverse-/forward-KL branch switch with a single mode-consistent reverse-KL distillation gated by a continuous attenuation weight, and prove the resulting per-edge objective is C^1 across the trust boundary while reverse-KL distillation is variance-stable and the forward-KL fallback provably inflates policy entropy in proportion to the mismatch; (ii) defines a scale-invariant, self-stabilizing trust radius from an agent's own policy nonstationarity, correcting the sign and scale pathologies of a value-variance radius. On the theory side we prove a joint *improvement lower bound* for simultaneous trust-region updates against frozen snapshots, making the cost of removing a sequential update queue explicit as an $O(N\sqrt{\eta})$ simultaneity penalty—a bound of the same family as the decentralized-PPO analysis of Sun et al. [7], here derived in the peer-distillation setting—and, under a standard exponential-decay-of-correlations assumption for networked systems, upgrade this to a per-capita bound whose error is independent of the swarm size N. We give a Byzantine-robust extension with a bounded-influence guarantee. This document specifies the mathematics only; empirical validation is specified in a companion protocol.

1 Introduction

In fully decentralized MARL, agents $i \in V = \{1, \ldots, N\}$ interact over a network $G = (V, E)$ with no central critic and no shared replay buffer. To learn faster than in isolation, each agent exchanges compact policy snapshots with its neighbors and distills their behavior into its own policy. This works when neighbors are behaviorally similar, but in heterogeneous swarms—agents facing different terrain, sensing limits, or roles—peer policies diverge, and distilling a divergent peer

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imports gradients that destroy an agent's local competence. The design problem is therefore: *how can each agent decide, online and using only local information, how much to trust each neighbor, so that useful knowledge flows while destabilizing knowledge is attenuated—without sacrificing a joint improvement guarantee?*

The conflation at the heart of the naive design. A direct decentralized transfer of trust-region distillation places a radius on the *social* divergence $\mathrm{KL}(\pi_i \,\|\, \pi_j)$ and then claims a monotonic-improvement bound. But the only divergence that appears in the trust-region improvement theorems of [1, 2, 4] is the *update* divergence $\mathrm{KL}(\pi_i^{\mathrm{new}} \,\|\, \pi_i^{\mathrm{old}})$ between an agent's successive policies. These are different quantities; bounding the social divergence to a neighbor says nothing about how far an agent moves. Recognizing and repairing this conflation is the organizing principle of this paper.

Contributions.

1. **Two trust regions** (Sec. 4). We separate the policy-improvement trust region from the distillation trust region and show (Remark 1) why constraining only the latter cannot yield an improvement guarantee.

2. **Continuous, mode-consistent distillation** (Secs. 5, 6). We replace the switched reverse-/forward-KL rule with a single reverse-KL term gated by a continuous attenuation weight, prove the per-edge objective is C^1 across the trust boundary (Prop. 2), and prove reverse-KL distillation is variance-stable while the forward-KL fallback inflates entropy in proportion to the mismatch (Prop. 3), justifying the single-direction choice.

3. **Self-stabilizing, scale-invariant radius** (Sec. 5.2). We define the trust radius from an agent's own policy nonstationarity rather than from the variance of its value function, correcting a sign pathology (a degenerate critic would earn maximal trust) and a reward-scale dependence.

4. **Simultaneity bound in the distillation setting** (Sec. 7). We prove a joint improvement *lower bound* (Thm. 1) for simultaneous updates against frozen snapshots and make the price of dropping a sequential queue explicit as an $O(N\sqrt{\eta})$ penalty. Related simultaneous-update trust-region bounds were established for decentralized PPO by Sun et al. [7]; our contribution is the derivation in the peer-distillation setting and the accompanying β-safety condition (Prop. 4) coupling the distillation weight to the improvement radius.

5. **Size-independent bound under locality** (Sec. 8). Under a standard decay-of-correlations assumption we prove a per-capita improvement bound whose error term is independent of N (Thm. 2), giving a concrete theoretical reason to prefer decentralization over a centralized critic at scale.

6. **Robust extension** (Sec. 9). We give a Byzantine-robust aggregation with a bounded-influence guarantee (Prop. 5).

2 Related Work

Centralized and sequential trust-region MARL. MAPPO [5] stabilizes multi-agent training with a centralized critic that observes the global state; HAPPO [6] obtains a monotonic joint-improvement guarantee by updating agents *sequentially*, which does not match a swarm that acts in parallel. Most directly related to our simultaneity analysis, Sun et al. [7] prove trust-region

bounds for *simultaneously* updating decentralized policies under the non-stationarity induced by independent ratios, and show the trust region can be enforced by scaling ratio clipping with the number of agents. Theorem 1 should be read as the analogue of such bounds in our peer-distillation setting; the novelty of this paper lies not in the simultaneity bound per se but in the trust-gated distillation mechanism it certifies (Secs. 5–6) and in the locality upgrade (Thm. 2).

Networked and consensus-based decentralized MARL. Fully decentralized MARL over communication graphs was formalized by Zhang et al. [13] via consensus actor-critic updates. MATRPO [8] transforms the TRPO update into a distributed consensus optimization in which agents share only likelihood ratios with neighbors; it enforces a *joint* trust region by consensus, whereas D-TrOPD keeps updates fully local and mediates peer influence through gated distillation rather than consensus on a common objective. Chu et al. [9] stabilize networked MARL with a spatial discount factor that attenuates distant agents' influence—a fixed, topology-based dampening, in contrast to our adaptive, behavior-based (KL) dampening. Qu et al. [12] establish the exponential-decay-of-correlations framework for networked RL on which our Assumption 2 and Theorem 2 rest.

Policy distillation. Policy distillation originates in single-agent transfer [10]. Trust-region on-policy distillation (TrOPD) [11] gates distillation by a KL trust region but assumes a single, fixed teacher–student pair; moreover it originates in large-language-model post-training, where the teacher is a genuinely superior fixed model and the KL-direction machinery addresses a Monte-Carlo estimator problem that does not arise for the closed-form Gaussian divergences used in continuous control. D-TrOPD generalizes the trust-gated idea to a symmetric many-to-many setting where every agent is simultaneously teacher and student, and replaces the direction switch with continuous attenuation.

Robust aggregation. Byzantine-tolerant gradient aggregation (Krum, trimmed means, geometric medians) is standard in distributed learning [14]; we import these operators for the distillation direction and state the resulting bounded-influence guarantee in our setting.

3 Preliminaries

Networked cooperative Markov game. We consider a tuple $\mathcal{M} = \big(V, G, \mathcal{S}, \{\mathcal{A}_i\}_{i\in V}, P, \{r_i\}_{i\in V}, \gamma\big)$. Here $V = \{1, \dots, N\}$ is the set of agents; $G = (V, E)$ is an undirected interaction/communication graph with neighborhood $\mathcal{N}_i = \{j : (i, j) \in E\}$, degree $d_i = |\mathcal{N}_i|$, and maximum degree $D_{\max} = \max_i d_i$. The (global) state is $s \in \mathcal{S}$; agent i observes $o_i = O_i(s)$ and selects $a_i \in \mathcal{A}_i$; the joint action is $a = (a_1, \dots, a_N)$. Transitions follow $P(s' \mid s, a)$ and $\gamma \in [0, 1)$ is the discount. The team reward is the average

$$r(s, a) = \frac{1}{N} \sum_{i\in V} r_i(s, a), \tag{1}$$

so that the objective is a per-capita quantity (this normalization is what makes the size-independent bound of Sec. 8 meaningful).

Decentralized policies. Each agent carries a parametric policy $\pi_i(a_i \mid o_i; \theta_i)$; the joint policy factorizes,

$$\pi_\theta(a \mid s) = \prod_{i\in V} \pi_i(a_i \mid o_i; \theta_i), \qquad \theta = (\theta_1, \dots, \theta_N). \tag{2}$$

For continuous control we take each π_i to be a tanh-squashed diagonal Gaussian: latent $u \sim \mathcal{N}\big(\mu_i(o_i), \mathrm{diag}(\sigma_i(o_i)^2)\big)$ and action $a_i = \tanh(u)$. Because tanh is a diffeomorphism applied identically to two policies, the KL divergence between squashed policies equals the KL between their latent Gaussians (Prop. 1); all divergences below are therefore computed in closed form on the latent parameters (μ, σ).

Value functions and visitation. For a joint policy π, let $V_\pi(s) = \mathbb{E}_\pi[\sum_{t\geq 0} \gamma^t r(s_t, a_t) \mid s_0 = s]$, $Q_\pi(s,a)$ the state–action value, and $A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$ the advantage. The (normalized) discounted state-visitation distribution is $d_\pi(s) = (1-\gamma)\sum_{t\geq 0}\gamma^t \Pr_\pi(s_t = s)$, and $J(\pi) = \mathbb{E}_{s\sim\rho_0}[V_\pi(s)]$ is the objective. We write $\varepsilon_\pi = \max_s \left|\mathbb{E}_{a\sim\pi} A_{\pi_{\mathrm{old}}}(s,a)\right|$ for the maximum expected advantage encountered under π.

Two divergence quantities. For a fixed reference distribution d over states we abbreviate the expected KL between two policies as $\bar{D}^d_{\mathrm{KL}}(\pi, \pi') := \mathbb{E}_{s\sim d}\big[\mathrm{KL}(\pi(\cdot \mid s) \,\|\, \pi'(\cdot \mid s))\big]$, and analogously $\bar{D}^d_{\mathrm{TV}}$ for total variation. Throughout, we distinguish:

Update divergence (per agent)	$\bar{D}^{d_\pi}_{\mathrm{KL}}\big(\pi_i^{(t)}, \pi_i^{(t+1)}\big)$
Social divergence (per edge)	$\bar{D}^{d}_{\mathrm{KL}}\big(\pi_i, \pi_j\big)$

4 The Two Trust Regions

We make the organizing distinction precise.

Definition 1 (Policy-improvement trust region)**.** *Agent i's update $\theta_i^{(t)} \to \theta_i^{(t+1)}$ lies in the* improvement trust region *of radius η_i if $\bar{D}^{d_{\pi^{(t)}}}_{\mathrm{KL}}\big(\pi_i^{(t)}, \pi_i^{(t+1)}\big) \leq \eta_i$.*

Definition 2 (Distillation trust region)**.** *Neighbor j is inside agent i's* distillation trust region *of radius δ_i if the social divergence $D_{ij} := \bar{D}^{d_{\pi_i}}_{\mathrm{KL}}\big(\pi_i, \pi_j\big)$ satisfies $D_{ij} \leq \delta_i$.*

The improvement trust region controls improvement guarantees (Sec. 7); the distillation trust region controls how much a peer is imported (Sec. 5).

Remark 1 (Why constraining only δ_i is insufficient). The trust-region improvement theorems of [1, 2, 4] bound $J(\pi') - J(\pi)$ by a surrogate minus a penalty in the *update* divergence $\bar{D}_{\mathrm{KL}}(\pi^{(t)}, \pi^{(t+1)})$. A radius δ_i on the social divergence D_{ij} places no constraint on how far $\pi_i^{(t+1)}$ moves from $\pi_i^{(t)}$: an agent can satisfy $D_{ij} \leq \delta_i$ for all j while taking an arbitrarily large update step (e.g., if the distillation gradient is large but the clip is inactive). Hence an improvement claim that cites only a social-divergence radius is unsupported. D-TrOPD therefore enforces *both* regions: δ_i shapes the social term, and a separate mechanism (PPO clipping with KL early stopping, Sec. 5.3) enforces η_i.

Crucially, distillation is off-policy with respect to π_i (it pulls toward a neighbor evaluated on states π_i may not visit), while the policy-gradient term is on-policy. To keep both faithful, the social divergence in Definition 2 is evaluated under π_i's own visitation d_{π_i}, not under a pooled batch; a pooled batch requires an importance correction (Remark 9).

5 The D-TrOPD Objective

5.1 Social distillation with continuous attenuation

Each agent distills from its neighbors using a single divergence direction—the reverse KL $\mathrm{KL}(\pi_i \,\|\, \pi_j)$, in which the student π_i is the first argument—weighted by a continuous attenuation coefficient

$w_{ij} \in (0, 1]$:

$$\mathcal{L}_i^{\text{soc}}(\theta_i) = \sum_{j \in \mathcal{N}_k(i)} w_{ij}\, \bar{D}_{\text{KL}}^{d_{\pi_i}}\big(\pi_{\theta_i}, \pi_{\theta_j^{(t)}}\big), \qquad w_{ij} = \exp\Big(-\gamma\big[\text{softplus}(D_{ij} - \delta_i) - \text{softplus}(-\delta_i)\big]\Big). \tag{3}$$

Here $\theta_j^{(t)}$ is a frozen snapshot of neighbor j at the start of epoch t (gradients through teachers are detached), $\mathcal{N}_k(i) \subseteq \mathcal{N}_i$ is a selected subset of neighbors (see Remark 2), $\gamma > 0$ is the attenuation rate, and $\text{softplus}(x) = \log(1 + e^x)$. The subtraction of $\text{softplus}(-\delta_i)$ normalizes the weight so that $w_{ij} = 1$ exactly at $D_{ij} = 0$ (a behaviorally identical neighbor is fully trusted); without it, $w_{ij} < 1$ even at zero divergence. The normalization is a constant offset in the exponent and does not affect smoothness.

Equation (3) replaces the discontinuous "reverse-KL inside / forward-KL outside" switch of a direct TrOPD transfer with a graceful, always-reverse-KL rule: inside the trust region $w_{ij} \approx 1$ (full distillation); as D_{ij} exceeds δ_i the weight decays smoothly to 0 (isolation). The softplus argument makes the exponent, and hence w_{ij}, continuously differentiable in D_{ij} everywhere including at $D_{ij} = \delta_i$ (Prop. 2). We show in Sec. 6 that retaining reverse KL—rather than switching to forward KL for distant peers—is the variance-stable choice, so the single-direction design is not merely simpler but strictly safer.

Remark 2 (Neighbor selection and smoothness). A hard top-k rule ($\mathcal{N}_k(i)$ = the k smallest-D_{ij} neighbors) reintroduces, at the selection boundary, exactly the discontinuity that Eq. (3) eliminates at the trust boundary: the total social loss jumps whenever a neighbor crosses into or out of the top-k set. Since the attenuation weight already suppresses distant neighbors exponentially, we recommend either (a) taking $\mathcal{N}_k(i) = \mathcal{N}_i$ (no selection; the weights do the filtering), or (b) a smooth soft-selection in which each neighbor's weight is further multiplied by a differentiable rank score. Under (a), the full objective inherits the C^1 property of Prop. 2; under a hard top-k, Prop. 2 holds per edge only. Where a hard cap on communication cost is required, top-k may be retained with this caveat stated.

5.2 A self-stabilizing, scale-invariant trust radius

The radius should shrink when an agent is itself unstable (so it does not import further perturbation) and should be invariant to reward scaling. We therefore set it from the agent's *own policy nonstationarity*, measured in nats:

$$\delta_i^{(t)} = \delta_{\text{base}} \cdot \exp\big(-\kappa\, \Delta_i^{(t)}\big), \qquad \Delta_i^{(t)} = \bar{D}_{\text{KL}}^{d_{\pi_i^{(t)}}}\big(\pi_i^{(t)}, \pi_i^{(t-1)}\big). \tag{4}$$

$\Delta_i^{(t)}$ is exactly the update divergence of the previous epoch: if the agent's own policy is churning, $\delta_i^{(t)}$ contracts and even moderate peers fall outside the trust region. Because $\Delta_i^{(t)}$ is a KL between policies (dimensionless in nats), the radius is invariant to affine reward rescaling and to the return scale $1/(1-\gamma)$, unlike a radius built from $\text{Var}_s(V_{\theta_i}(s))$.

Remark 3 (Why not the value-function variance). A radius $\delta_i \propto \exp(-\kappa \text{Var}_s V_{\theta_i}(s))$ measures the spread of values across states, which is governed by reward heterogeneity, not by instability. A degenerate critic with a near-constant value head has $\text{Var}_s V \approx 0$ and would earn the *maximal* radius (trust everyone), whereas a well-trained critic in a reward-differentiated task has large $\text{Var}_s V$ and would be isolated—the opposite of the intent, and additionally non-invariant to reward scale. Equation (4) removes both pathologies. An alternative instability signal with the same invariance is the critic's explained variance $\text{EV}_i = 1 - \text{Var}(G_i - V_{\theta_i})/\text{Var}(G_i)$, where G_i are empirical returns;

one may set $\delta_i = \delta_{\text{base}}\,\varsigma(a(\text{EV}_i - b))$ for a logistic ς. We use (4) as it reuses the KL machinery already present.

Remark 4 (Effective dynamic range of the radius). Because KL early stopping (Sec. 5.3) caps each epoch's update divergence at η_i, the signal $\Delta_i^{(t)}$ lives in $[0, \eta_i]$ and hence $\delta_i^{(t)} \in [\delta_{\text{base}} e^{-\kappa\eta_i}, \delta_{\text{base}}]$. The sensitivity κ should therefore be chosen relative to η_i (e.g., $\kappa\eta_i \in [1, 3]$) so that the radius has meaningful dynamic range; with $\kappa\eta_i \ll 1$ the radius is effectively constant.

5.3 Combined objective and both trust regions

Agent i maximizes its clipped policy-gradient surrogate minus the attenuated social penalty,

$$\mathcal{L}_i(\theta_i) = \mathcal{L}_i^{\text{CLIP}}(\theta_i) - \beta\,\mathcal{L}_i^{\text{soc}}(\theta_i), \tag{5}$$

with the standard PPO surrogate on agent i's own rollout $\{(s_\tau^i, a_\tau^i)\}$,

$$\mathcal{L}_i^{\text{CLIP}}(\theta_i) = \mathbb{E}_\tau\Big[\min\big(\rho_\tau^i \hat{A}_\tau^i,\ \text{clip}(\rho_\tau^i, 1-\epsilon, 1+\epsilon)\,\hat{A}_\tau^i\big)\Big], \qquad \rho_\tau^i = \frac{\pi_{\theta_i}(a_\tau^i \mid o_\tau^i)}{\pi_{\theta_i^{\text{old}}}(a_\tau^i \mid o_\tau^i)}. \tag{6}$$

The two trust regions enter as follows. The distillation region is imposed by w_{ij} and δ_i in (3). The improvement region η_i of Definition 1 is imposed operationally by (a) the clip in (6) and (b) *KL early stopping*: inner optimization of (5) halts once $\bar{D}_{\text{KL}}^{d_{\pi_i^{(t)}}}(\pi_i^{(t)}, \pi_{\theta_i})$ reaches η_i. A critic loss $\mathcal{L}_i^V = \mathbb{E}_\tau[(V_{\theta_i}(s_\tau^i) - \hat{G}_\tau^i)^2]$ is optimized jointly so that advantages and the radius signal (4) are computed from a trained value function.[1]

5.4 Algorithm

Algorithm 1 exchanges only policy snapshots over edges of G (never raw observations or gradients), uses only i's own rollout for the on-policy term, and freezes teachers for the epoch so that no agent chases a moving target. All communication is neighbor-local; there is no global batch. We next verify the properties this design was built to guarantee.

6 Structural Properties of the Objective

Proposition 1 (Latent-space KL is exact)**.** *Let π_i, π_j be tanh-squashed diagonal Gaussians with latents $\mathcal{N}(\mu_i, \Sigma_i)$, $\mathcal{N}(\mu_j, \Sigma_j)$. Then* $\text{KL}(\pi_i(\cdot \mid s) \,\|\, \pi_j(\cdot \mid s)) = \text{KL}(\mathcal{N}(\mu_i, \Sigma_i) \,\|\, \mathcal{N}(\mu_j, \Sigma_j))$.

Proof. $\tanh : \mathbb{R}^m \to (-1, 1)^m$ is a C^∞ diffeomorphism. For a common invertible, differentiable map g applied to both densities, the Jacobian factors $|\det \nabla g|$ cancel inside the log-ratio, so KL is invariant: $\text{KL}(g_\# p \,\|\, g_\# q) = \text{KL}(p \,\|\, q)$. Apply with $g = \tanh$. □

Proposition 2 (C^1 per-edge objective; no chattering across the trust boundary)**.** *Fix an edge (i, j) and view the per-edge penalty as a function of the social divergence, $g_{ij}(D) = \exp\big(-\gamma[\text{softplus}(D - \delta_i) - \text{softplus}(-\delta_i)]\big) \cdot D$ for $D \geq 0$. Then g_{ij} is continuously differentiable on $[0, \infty)$; in particular it has no jump in value or first derivative at $D = \delta_i$.*

[1] We flag this explicitly because it is the one place where an implementation must not silently drop the value-regression term: the radius (4) and the advantages $\hat{A}^i$ both depend on a critic that is actually optimized toward returns.

Algorithm 1 D-TrOPD (one epoch t, executed identically and in parallel by every agent i)

Input: snapshots $\{\theta_j^{(t)}\}_{j\in\mathcal{N}_i}$ received from neighbors; params $\delta_{\text{base}}, \kappa, \gamma, \beta$; trust radius η_i.
Broadcast a frozen copy of $\theta_i^{(t)}$ to neighbors $\mathcal{N}_i$; freeze received snapshots (detach).
Roll out $\pi_{\theta_i^{(t)}}$ locally; collect $\{(o_\tau^i, a_\tau^i, r_\tau^i)\}$ and estimate $\hat{A}^i$ (GAE) and returns $\hat{G}^i$.
Compute own nonstationarity $\Delta_i^{(t)}$ (Eq. 4) and set $\delta_i^{(t)}$.
for each neighbor $j \in \mathcal{N}_i$ **do**
    $D_{ij} \leftarrow \bar{D}_{\text{KL}}^{d_{\pi_i}}(\pi_{\theta_i^{(t)}}, \pi_{\theta_j^{(t)}})$ ▷ closed-form Gaussian KL on own states
    $w_{ij} \leftarrow \exp(-\gamma[\text{softplus}(D_{ij} - \delta_i^{(t)}) - \text{softplus}(-\delta_i^{(t)})])$
end for
$\mathcal{N}_k(i) \leftarrow \mathcal{N}_i$ (default; see Remark 2 for capped variants).
repeat ▷ inner optimization with both trust regions
    $\mathcal{L}_i \leftarrow \mathcal{L}_i^{\text{CLIP}}(\theta_i) - \beta \sum_{j\in\mathcal{N}_k(i)} w_{ij}\, \bar{D}_{\text{KL}}^{d_{\pi_i}}(\pi_{\theta_i}, \pi_{\theta_j^{(t)}}) - c_V\, \mathcal{L}_i^V(\theta_i)$
    $\theta_i \leftarrow \theta_i + \alpha\, \nabla_{\theta_i} \mathcal{L}_i$
until $\bar{D}_{\text{KL}}^{d_{\pi_i^{(t)}}}(\pi_i^{(t)}, \pi_{\theta_i}) \geq \eta_i$ **or** max inner steps
$\theta_i^{(t+1)} \leftarrow \theta_i$.

Proof. softplus is C^∞, hence $w(D) = \exp(-\gamma[\text{softplus}(D-\delta_i) - \text{softplus}(-\delta_i)])$ is C^∞, and $g_{ij} = w \cdot D$ is a product of C^∞ functions, so $g_{ij} \in C^\infty \subset C^1$. □

Proposition 2 is the formal counterpart of "graceful, not binary": a switched objective $g(D) = D \cdot \mathbf{1}[D \leq \delta] + \psi(D)\,\text{KL}(\pi_j \,\|\, \pi_i) \cdot \mathbf{1}[D > \delta]$ is discontinuous at $D = \delta$ whenever $\text{KL}(\pi_i \,\|\, \pi_j) \neq \text{KL}(\pi_j \,\|\, \pi_i)$ there (generic for Gaussians with $\Sigma_i \neq \Sigma_j$), producing sign-flipping updates as edges drift across a moving boundary. Equation (3) removes this. Note the scope: with hard top-k selection, smoothness holds per edge but not for the summed objective (Remark 2).

Proposition 3 (Reverse-KL is variance-stable; forward-KL inflates entropy)**.** *Let teacher $\pi_j = \mathcal{N}(\mu_j, \sigma_j^2)$ be fixed (per dimension) and optimize the student $\pi_i = \mathcal{N}(\mu_i, \sigma_i^2)$.*

(a) Minimizing the reverse KL $\text{KL}(\pi_i \,\|\, \pi_j)$ over (μ_i, σ_i) has the unique stationary point $\mu_i^\star = \mu_j$, $\sigma_i^\star = \sigma_j$. The optimal student variance is independent of the mean gap.

(b) Minimizing the forward KL $\text{KL}(\pi_j \,\|\, \pi_i)$ over (μ_i, σ_i) has, at any fixed μ_i, the stationary condition $(\sigma_i^\star)^2 = \sigma_j^2 + (\mu_i - \mu_j)^2$; in particular, away from $\mu_i = \mu_j$ the forward-KL gradient increases σ_i in proportion to the squared mean gap.

Proof. Per dimension, $\text{KL}(\pi_i \,\|\, \pi_j) = \log \frac{\sigma_j}{\sigma_i} + \frac{\sigma_i^2 + (\mu_i - \mu_j)^2}{2\sigma_j^2} - \frac{1}{2}$. Then $\partial_{\mu_i} = \frac{\mu_i - \mu_j}{\sigma_j^2}$ and $\partial_{\sigma_i} = -\frac{1}{\sigma_i} + \frac{\sigma_i}{\sigma_j^2}$, giving $\mu_i^\star = \mu_j$, $\sigma_i^\star = \sigma_j$. The Hessian is positive definite, so the stationary point is the unique minimizer. This proves (a). For (b), $\text{KL}(\pi_j \,\|\, \pi_i) = \log \frac{\sigma_i}{\sigma_j} + \frac{\sigma_j^2 + (\mu_i - \mu_j)^2}{2\sigma_i^2} - \frac{1}{2}$, so $\partial_{\sigma_i} = \frac{1}{\sigma_i} - \frac{\sigma_j^2 + (\mu_i - \mu_j)^2}{\sigma_i^3}$, which vanishes at $(\sigma_i^\star)^2 = \sigma_j^2 + (\mu_i - \mu_j)^2$; for $\sigma_i < \sigma_i^\star$ the derivative is negative, i.e., gradient descent increases σ_i, and the target grows with $(\mu_i - \mu_j)^2$. □

Remark 5 (Consequence for the discarded fallback). The distillation trust region is exited precisely when D_{ij} is large, which for Gaussians is driven by a large mean gap $(\mu_i - \mu_j)^2/2\sigma^2$. By Prop. 3(b), a forward-KL fallback there pulls the student's variance toward $\sigma_j^2 + (\mu_i - \mu_j)^2$, i.e., inflates the student's entropy in proportion to the mismatch that triggered the fallback—the exact "policy

destruction under distribution mismatch" the mechanism is meant to prevent. Reverse KL instead leaves the variance target at σ_j. This is why D-TrOPD keeps reverse KL for all neighbors and attenuates by weight, not by direction.

Proposition 4 (β-safety: distillation stays inside the improvement region)**.** *Assume each policy is L_π-smooth in θ_i so that, to second order, $\bar{D}_{\mathrm{KL}}^{d_{\pi_i}}(\pi_{\theta_i^{(t)}}, \pi_{\theta_i^{(t)}+\Delta\theta}) \le \frac{1}{2}F_{\max}\|\Delta\theta\|^2$ with $F_{\max}$ an upper bound on the Fisher/Hessian spectrum, and that per-edge social gradients are bounded, $\|\nabla_{\theta_i}\bar{D}_{\mathrm{KL}}(\pi_{\theta_i}, \pi_{\theta_j^{(t)}})\| \le G$. For a single ascent step $\Delta\theta = \alpha\nabla_{\theta_i}\mathcal{L}_i$, the update lies in the improvement trust region of radius η_i whenever*

$$\beta \;\le\; \beta_{\max} := \frac{1}{kG}\left(\frac{\sqrt{2\eta_i/F_{\max}}}{\alpha} - \left\|\nabla_{\theta_i}\mathcal{L}_i^{\mathrm{CLIP}}\right\|\right), \tag{7}$$

provided the right-hand side is positive, where $k = |\mathcal{N}_k(i)|$.

Proof. By the triangle inequality $\|\nabla\mathcal{L}_i\| \le \|\nabla\mathcal{L}_i^{\mathrm{CLIP}}\| + \beta\sum_{j\in\mathcal{N}_k(i)} w_{ij}\|\nabla\bar{D}_{\mathrm{KL}}\| \le \|\nabla\mathcal{L}_i^{\mathrm{CLIP}}\| + \beta kG$ since $w_{ij} \le 1$ and $|\mathcal{N}_k(i)| \le k$. The smoothness bound gives the update KL $\le \frac{1}{2}F_{\max}\alpha^2\|\nabla\mathcal{L}_i\|^2$. Requiring this $\le \eta_i$ yields $\alpha\|\nabla\mathcal{L}_i\| \le \sqrt{2\eta_i/F_{\max}}$; substituting the gradient-norm bound and solving for β gives (7). □

Proposition 4 ties the social weight β to the improvement radius η_i: the distillation term is guaranteed not to push a single update step out of the region that governs the improvement bound. This is the quantitative form of "both trust regions hold simultaneously." (The statement is per-step; over multiple inner steps, KL early stopping provides the operative guarantee.)

7 An Improvement Lower Bound under Simultaneous Updates

We now prove a joint improvement lower bound without a sequential update queue, and expose the price paid for parallelism. Bounds of this family for simultaneously updating decentralized policies were first established by Sun et al. [7]; we state and prove the version arising in our setting, where updates are taken against frozen snapshots and the improvement region is enforced by KL early stopping, both because our constants differ and to keep the paper self-contained. We use the following standard ingredients.

Assumption 1 (Regularity)**.** *(i) Policies factorize as in (2). (ii) The maximum expected advantage is bounded, $\varepsilon_\pi \le \varepsilon$ for all iterates. (iii) Each agent's update lies in its improvement trust region, $\bar{D}_{\mathrm{KL}}^{d_{\pi^{(t)}}}(\pi_i^{(t)}, \pi_i^{(t+1)}) \le \eta_i$ (enforced by clipping and KL early stopping, Sec. 5.3).*

Lemma 1 (Additivity of joint divergence)**.** *For product policies (2) and every state s,*

$$\mathrm{KL}\big(\pi(\cdot\mid s)\,\big\|\,\pi'(\cdot\mid s)\big) = \sum_{i\in V}\mathrm{KL}\big(\pi_i(\cdot\mid o_i)\,\big\|\,\pi_i'(\cdot\mid o_i)\big), \qquad \mathrm{TV}\big(\pi(\cdot\mid s),\pi'(\cdot\mid s)\big) \le \sum_{i\in V}\mathrm{TV}\big(\pi_i(\cdot\mid o_i),\pi_i'(\cdot\mid o_i)\big). \tag{8}$$

Proof. For the KL, substitute $\pi = \prod_i \pi_i$, $\pi' = \prod_i \pi_i'$ into the definition; the logarithm of the product splits into a sum and each cross term integrates to 1, leaving $\sum_i \mathrm{KL}(\pi_i \,\|\, \pi_i')$. For the TV, use the coupling/telescoping bound: replacing one coordinate's marginal at a time and applying the triangle inequality for TV of product measures gives subadditivity $\mathrm{TV}(\prod_i \pi_i, \prod_i \pi_i') \le \sum_i \mathrm{TV}(\pi_i, \pi_i')$. □

Lemma 2 (Surrogate lower bound [4, 2, 1])**.** *For any joint policies π, π',*

$$J(\pi') \geq L_\pi(\pi') - \frac{2\gamma\varepsilon}{(1-\gamma)^2}\,\bar{D}_{\mathrm{TV}}^{d_\pi}(\pi, \pi'), \qquad L_\pi(\pi') := J(\pi) + \frac{1}{1-\gamma}\,\mathbb{E}_{\substack{s\sim d_\pi \\ a\sim\pi'}}\big[A_\pi(s,a)\big]. \tag{9}$$

Theorem 1 (Joint improvement lower bound with an explicit simultaneity penalty)**.** *Under Assumption 1, a single round of simultaneous updates satisfies*

$$J(\pi^{(t+1)}) \geq L_{\pi^{(t)}}(\pi^{(t+1)}) - \frac{2\gamma\varepsilon}{(1-\gamma)^2}\sum_{i\in V}\sqrt{\frac{\eta_i}{2}}. \tag{10}$$

In particular, if $\eta_i = \eta$ for all i, the penalty equals $\frac{2\gamma\varepsilon}{(1-\gamma)^2}\cdot N\sqrt{\eta/2}$, i.e., it grows linearly in the number of agents N.

Proof. Start from Lemma 2. By Lemma 1 and Pinsker's inequality $\mathrm{TV}(p,q) \leq \sqrt{\frac{1}{2}\mathrm{KL}(p\,\|\,q)}$ applied per agent, then Jensen's inequality,

$$\bar{D}_{\mathrm{TV}}^{d_{\pi^{(t)}}}(\pi^{(t)}, \pi^{(t+1)}) \leq \sum_i \mathbb{E}_s\big[\mathrm{TV}(\pi_i^{(t)}, \pi_i^{(t+1)})\big] \leq \sum_i \mathbb{E}_s\Big[\sqrt{\tfrac{1}{2}\mathrm{KL}(\pi_i^{(t)}\,\|\,\pi_i^{(t+1)})}\Big] \leq \sum_i \sqrt{\tfrac{1}{2}\eta_i},$$

where the last step uses concavity of $\sqrt{\cdot}$ and Assumption 1(iii). Substituting into (9) gives (10). □

Remark 6 (On the meaning of the bound). Theorem 1 is a *lower bound*, not by itself a monotonic-improvement statement: the return improves only when the surrogate gain $L_{\pi^{(t)}}(\pi^{(t+1)}) - J(\pi^{(t)})$ exceeds the simultaneity penalty. This is the standard reading of trust-region bounds [2, 4]; we adopt it explicitly to avoid overclaiming.

Remark 7 (The price of dropping the queue). HAPPO obtains a size-free monotonic-improvement guarantee by updating agents sequentially, so that the multi-agent advantage decomposition telescopes exactly and no cross-agent surrogate error remains. Theorem 1 shows what simultaneity costs: an additive $O(N\sqrt{\eta})$ penalty, which is exactly the surrogate error from all agents moving at once against frozen snapshots. (Sun et al. [7] manage the analogous penalty by scaling the ratio-clipping range with N.) This penalty is benign when steps are small ($\eta \to 0$) or when N is moderate, but it grows with the swarm—precisely the regime a swarm method targets. Section 8 removes the N-dependence under locality.

8 A Size-Independent Bound under Locality

The linear-in-N penalty of Theorem 1 treats every pair of agents as coupled. In a physical swarm, interactions are local: reward (1) and transitions factor over G, and the influence of a distant agent decays. We formalize this with the standard decay-of-correlations assumption of networked RL [12, 13].

Assumption 2 (Local dependence and exponential decay)**.** *(i) Agent i's reward r_i and the transition of its local state depend only on $(s_{\mathcal{N}_i}, a_{\mathcal{N}_i})$. (ii) There exist $c > 0$ and $\rho \in [0,1)$ such that for any two joint policies $\pi, \tilde{\pi}$ that agree on the κ-hop ball $B_\kappa(i) = \{j : \mathrm{dist}_G(i,j) \leq \kappa\}$,*

$$\big|Q_\pi^i(s,a) - Q_{\tilde{\pi}}^i(s,a)\big| \leq c\,\rho^\kappa, \tag{11}$$

where Q_π^i is the local action-value of the term r_i.

Theorem 2 (Per-capita improvement lower bound, independent of N)**.** *Under Assumptions 1 and 2, suppose each agent restricts its update to information within its κ-hop ball and stays within radius η. Then the per-capita return obeys*

$$\frac{1}{N}\Big[J(\pi^{(t+1)}) - J(\pi^{(t)})\Big] \geq \frac{1}{N}\sum_{i\in V}\ell_i\big(\pi_i^{(t+1)}\big) - C_{\mathrm{loc}}\,|B_\kappa|_{\max}\sqrt{\frac{\eta}{2}} - \frac{c\,\rho^\kappa}{1-\gamma}, \tag{12}$$

where $\ell_i(\pi_i') = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{\pi^{(t)}},\,a\sim\pi'}[A^i_{\pi^{(t)}}(s,a)]$ is agent i's local surrogate gain, $|B_\kappa|_{\max} \leq D^\kappa_{\max}$ is the largest κ-ball size, and $C_{\mathrm{loc}} = \frac{2\gamma\varepsilon}{(1-\gamma)^2}$. Both the coupling penalty $C_{\mathrm{loc}}|B_\kappa|_{\max}\sqrt{\eta/2}$ and the truncation error $c\rho^\kappa/(1-\gamma)$ are independent of N.

Proof sketch. Decompose the team advantage as $A_\pi = \frac{1}{N}\sum_i A^i_\pi$ by linearity of (1). Apply Lemma 2 to each local term A^i_π. By Assumption 2(ii), A^i_π is $c\rho^\kappa$-insensitive to policy changes outside $B_\kappa(i)$; hence in the surrogate error for term i we may replace the global policy change by the change restricted to $B_\kappa(i)$, incurring an additive $c\rho^\kappa/(1-\gamma)$. The restricted TV change is bounded, via Lemma 1 and Pinsker, by $\sum_{j\in B_\kappa(i)}\sqrt{\eta_j/2} \leq |B_\kappa|_{\max}\sqrt{\eta/2}$. Averaging over i (per-capita normalization) leaves the neighborhood-sized penalty and the truncation term, neither of which depends on N. Deferred details, including the truncated policy-gradient argument of [12] and the required visitation-mismatch bound, are outlined in Appendix A; a fully self-contained proof is left to the companion paper, and the theorem should be read with that caveat. □

Remark 8 (Interpretation). Theorem 2 is the theoretical payoff that distinguishes D-TrOPD from centralized-critic and sequential methods at scale: whereas MAPPO's centralized critic must observe all N agents and Theorem 1's naive penalty grows with N, a bounded interaction degree yields a per-capita improvement bound whose error is controlled by the local ball size $D^\kappa_{\max}$ and the decay rate ρ, and κ trades approximation error ρ^κ against locality $D^\kappa_{\max}$. The frozen-snapshot design of Algorithm 1 is exactly what makes the per-epoch update KL η the controlled quantity in (12).

Remark 9 (Pooled reference batch requires a correction). If, for efficiency, the social divergence D_{ij} is evaluated on a pooled batch $\bar{d}$ rather than d_{π_i}, then (12) no longer applies verbatim: the distilled quantity is a KL under $\bar{d}$, whereas the improvement bound is stated under d_{π_i}. The two are reconciled by importance weighting, $\bar{D}^{d_{\pi_i}}_{\mathrm{KL}} = \mathbb{E}_{s\sim\bar{d}}\big[\frac{d_{\pi_i}(s)}{\bar{d}(s)}\mathrm{KL}(\pi_i(\cdot\mid s)\,\|\,\pi_j(\cdot\mid s))\big]$, whose variance is controlled only when $d_{\pi_i} \ll \bar{d}$ with bounded ratio. We therefore evaluate D_{ij} on the agent's own rollout states by default.

9 Robustness to Byzantine Neighbors

A social mechanism that trusts nearby peers is exposed to adversarial peers. A KL trust test on a shared batch is spoofable: an adversary can appear close on the batch (low D_{ij}, hence $w_{ij} \approx 1$) while behaving adversarially on the victim's own states. Two safeguards follow directly from our formulation.

First, evaluating D_{ij} under d_{π_i} (Def. 2) already forces the trust test onto the states where distillation acts, closing the shared-batch spoof. Second, we aggregate neighbor distillation directions with a Byzantine-robust operator rather than a plain sum. Let $g_{ij} = \nabla_{\theta_i}\bar{D}_{\mathrm{KL}}(\pi_{\theta_i}, \pi_{\theta_j^{(t)}})$ be the per-neighbor distillation gradient, and let Agg be the coordinate-wise trimmed mean (or geometric median / Krum [14]) over $j \in \mathcal{N}_k(i)$.

Proposition 5 (Bounded influence of Byzantine neighbors)**.** *Suppose at most a fraction $f < \frac{1}{2}$ of an agent's k selected neighbors are Byzantine, and honest per-neighbor attenuated gradients $\{w_{ij}g_{ij}\}$*

are independent with mean $\bar{g}_w$ and coordinate-wise variance at most ς^2. Then the coordinate-wise f-trimmed mean $\hat{g} = \mathrm{Agg}(\{w_{ij} g_{ij}\})$ satisfies, in each coordinate,

$$|\hat{g} - \bar{g}_w| \le B(f, \varsigma, k),$$

where the bias B is finite for all $f < \frac{1}{2}$, is nonincreasing to the sampling error $O(\varsigma/\sqrt{k})$ as $f \to 0$, and does not depend on the magnitude of the Byzantine inputs. In particular the distilled update direction has bounded deviation from the honest attenuated mean whenever fewer than half of the selected neighbors are adversarial.

Proof sketch. Coordinate-wise, an f-trimmed mean discards the largest and smallest f-fractions of the inputs. Byzantine values that are extreme in a coordinate fall in the trimmed tails and are discarded; Byzantine values placed *inside* the honest range are not discarded, but their per-coordinate displacement of the trimmed mean is then bounded by the honest inter-quantile spread, which is $O(\varsigma)$ under the bounded-variance assumption. Discarding at most an f-fraction of honest values perturbs the honest mean by an amount controlled by the same spread. Combining the two effects gives a bias $B(f, \varsigma, k)$ that is finite for $f < \frac{1}{2}$ and vanishes (up to sampling error) as $f \to 0$; this is the standard robust-mean-estimation argument [14]. The attenuation weights $w_{ij} \in (0, 1]$ only rescale honest contributions and do not affect the breakdown fraction. □

Remark 10 (Scope of the robustness claim). Two honest caveats. (a) The independence assumption on honest gradients is an idealization: different neighbors induce different divergences and their gradients are correlated through the shared student. The qualitative conclusion—Byzantine influence is bounded by the honest spread rather than by the adversary's magnitude—survives, but sharp constants require the correlated analysis. (b) An adaptive adversary that stays inside the honest range can still bias the update by up to the honest spread; robustness here means *graceful degradation with a stated tolerance*, not immunity. Proposition 5 converts the adversarial-agent stressor from an uncontrolled hazard into a mechanism with a breakdown point of one half of an agent's trusted neighborhood.

10 Discussion

What the theory buys, and what it assumes. The results are organized so that assumptions are visible. Propositions 1–4 and Theorem 1 are self-contained given the regularity Assumption 1 and the cited single-agent bound (Lemma 2). The size-independent Theorem 2 rests additionally on the decay-of-correlations Assumption 2, which is standard for networked systems but should be tested empirically (e.g., by measuring how quickly D_{ij}-sensitivity falls with graph distance), and its proof is completed only in outline here. The Byzantine guarantee assumes fewer-than-half-adversarial trusted neighborhoods and idealized honest-gradient independence.

Design choices, justified rather than asserted. Three choices in the original design are replaced by ones the theory supports. (1) Single reverse-KL rather than a direction switch: Prop. 3 and Remark 5 show the forward-KL fallback inflates entropy exactly under the mismatch it is invoked for, whereas continuous attenuation (Prop. 2) delivers the intended graceful isolation without a discontinuity. (2) Self-nonstationarity radius rather than value-variance radius: Remark 3 shows the value-variance radius has the wrong sign and is reward-scale dependent. (3) Two explicit trust regions rather than one: Remark 1 shows a social-divergence radius cannot by itself support an improvement claim, and Prop. 4 shows how β is bounded so both regions hold at once.

Alternative geometry. Replacing KL with the 2-Wasserstein distance between the latent Gaussians, $W_2^2(\pi_i, \pi_j) = \|\mu_i - \mu_j\|^2 + \|\sigma_i - \sigma_j\|^2$, is symmetric (no direction to choose), finite, and a metric—so the triangle inequality would let one bound divergence along multi-hop paths directly, sharpening the "multi-hop influence dampening" intuition. The trade-off is that classical improvement bounds are stated in KL/TV, not W_2; committing to W_2 would move the analysis into the Wasserstein-natural-gradient setting. We flag this as the most promising route to a geometry-native multi-hop bound.

Scope. This document specifies mathematics only. A companion paper must (i) implement the trained critic and both trust regions, (ii) verify Theorem 2's decay assumption and the $O(N\sqrt{\eta})$ vs. size-independent penalties empirically, (iii) report multi-seed results against MAPPO, HAPPO, and IPPO with confidence intervals, and (iv) ablate the attenuation, the radius signal, and the robust aggregator. Earlier empirical results obtained under the superseded design (value-variance radius; KL-direction switch; pooled-batch trust test) do not bear on the corrected mechanism and are not reported.

11 Conclusion

We have given a theoretical blueprint for decentralized trust-region policy distillation that resolves a conflation between the policy-improvement and distillation trust regions, replaces a discontinuous KL-direction switch with a provably variance-stable and C^1 attenuation, grounds the trust radius in a scale-invariant instability signal, and proves a joint improvement lower bound for simultaneous updates whose error is independent of swarm size under a standard locality assumption. Together these turn a set of heuristics into statements with explicit hypotheses, and they identify the locality bound (Theorem 2) as the place where a decentralized method can provably beat a centralized critic at scale.

Appendix A: Deferred details for Theorem 2

The truncation step uses the exponential-decay property (11) together with a truncated policy-gradient argument in the style of [12]: the true local surrogate gradient $\nabla_{\theta_i}\ell_i$ differs from its κ-hop truncation by at most $O(\rho^\kappa)$, so restricting each agent's update to $B_\kappa(i)$ perturbs the per-term surrogate by at most $c\rho^\kappa/(1-\gamma)$. The neighborhood TV bound follows by applying Lemma 1 on $B_\kappa(i)$ instead of V; summing the N local terms and dividing by N yields (12). A complete proof additionally requires a bound on the mismatch between $d_{\pi^{(t)}}$ and $d_{\pi^{(t+1)}}$ restricted to $B_\kappa(i)$, which is $O(\sqrt{\eta}\,|B_\kappa|_{\max})$ by the simulation lemma; the constants are absorbed into C_{loc}. The fully self-contained proof, with all constants tracked, is deferred to the companion paper.

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